

COMPOSITE MATERIAL DATASHEETS CFRP HiTape/RTM6-UD (V100)

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	HiTape/RTM6-UD
Commercial name	Hexcel HiTape UD210 IMA/V100/ZD7/12.7MM with RTM6
Specifications	(*) This material is not qualified according Airbus standards and specifications. Data herein shall be consider preliminary until the qualification and further tests programs planned are concluded
Description	Carbon fiber uni-directional fabric in epoxy resin. Produced by a liquid resin infusion (LRI) process. Service temperature: -55 to 90°C. Cure: 120-135 minutes at 185 +/-5°C.
Version	0.5 (11/01/2024)

REFERENCES

[1] TE-T-NT-230075 Is.A "DERIVATION OF DESIGN VALUES FOR CFRP HiTape/RTM6-UD (V100) MATERIAL"

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.196 mm
Density	1540 kg/m ³
Tg-onset (wet)	175 °C
Max. Operational Limit (MOL)	TBD °C

Coefficients of Thermal Expansion (1/°C)	Range	-55..20°C	20..70°C	70..120°C
	α_1	0.0	0.0	0.0
	α_2	3.0E-5	3.0E-5	3.0E-5
	α_3	3.0E-5	3.0E-5	3.0E-5

ELASTIC MODULI

Property	E_1	E_2	G_{12}	ν_{12}	K_B
Units	MPa	MPa	MPa	-	-
Values	157000	6900	3300	0.4	0.88

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1 , E_2 and G_{12} in buckling stability analyses
- Apparent modulus of the quasi-isotropic laminate $E_{qi} = 57334$ MPa
- Estimated ν_{23} truncated

Property	E_3	G_{13}	G_{23}	ν_{13}	ν_{23}
Units	MPa	MPa	MPa	-	-
Values	6900	3300	1900	0.4	0.5

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	-55C/Dry	RT/Dry	70°C/W	90°C/W		-55C/Dry	70°C/W	90°C/W	-55C/Dry	RT/Dry	70°C/W	90°C/W
F_{1t}	2712	2806	2704	2501	0.900	0.966	0.964	0.891	2441	2526	2434	2251
F_{1c}	1835	1526	928	863.5	0.774	1	0.608	0.566	1420	1181	718.3	668.3
F_{2t}	-	46.9	27.3	19.8	0.760	-	0.582	0.422	-	35.6	20.7	15.0
F_{2c}	-	200.5	121.7	99.2	0.768	-	0.607	0.495	-	154.0	93.5	76.2
F_{12}	92.4	89.4	59.1	49.1	0.805	1	0.661	0.549	74.4	72.0	47.6	39.5

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values ($\mu\epsilon$)				B-basis K_B	E-KDF			B-basis Values ($\mu\epsilon$)			
	-55C/Dry	RT/Dry	70°C/W	90°C/W		-55C/Dry	70°C/W	90°C/W	-55C/Dry	RT/Dry	70°C/W	90°C/W
ϵ_{tall}	17000	17000	15400	13400	0.877	1	0.906	0.788	14900	14900	13500	11700
ϵ_{call}	8700	8700	4800	3600	0.780	1	0.552	0.414	6800	6800	3700	2800

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INTERLAMINAR STRENGTHS

Property	Mean Values				B-basis K_B	E-KDF			B-basis Values			
	-55C/Dry	RT/Dry	70°C/W	90°C/W		-55C/Dry	70°C/W	90°C/W	-55C/Dry	RT/Dry	70°C/W	90°C/W
F_{13} (MPa)	112.1	75.1	51.9	45.9	0.815	1	0.691	0.612	91.4	61.2	42.3	37.4
F_{3t} (MPa)	-	-	-	-	-	-	1	-	-	-	-	-
F_{3c} (MPa)	-	-	-	-	-	-	-	-	-	-	-	-
G_{IC} (N/mm)	0.78	0.79	0.79	0.79	0.815	0.994	1	1	0.64	0.64	0.64	0.64
G_{IIC} (N/mm)	1.34	1.40	0.72	0.51	0.753	0.957	0.514	0.367	1.0	1.1	0.5	0.4

Notes: F3t shall be made equal to F2t in QI and directed laminates if thickness $\geq 3\text{mm}$, otherwise contact Stress Methods dpt. Through-thickness compression strength F3c shall be made equal to F2c

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	-55C/Dry	RT/Dry	70°C/W	90°C/W		-55C/Dry	70°C/W	90°C/W	-55C/Dry	RT/Dry	70°C/W	90°C/W
OHT_{QI}	-	560.7	560.7	560.7	0.900	-	1	1	-	504.6	504.6	504.6
OHC_{QI}	-	277.7	208.0	175.4	0.834	-	0.749	0.632	-	231.6	173.5	146.3
FHT_{QI}	-	556.7	556.7	543.3	0.900	-	1	0.976	-	501	501	489
FHC_{QI}	-	380.4	274.5	240.9	0.834	-	0.722	0.633	-	317.3	228.9	200.9
F_{BR-QI}	1188	972.4	790.8	674.1	0.834	1	0.813	0.693	990.8	811.0	659.5	562.2
F_{PO}	-	-	-	-	-	-	-	-	-	-	-	-

Note: EKDF's for OH and FH strengths taken from plain-strength data

COMPOSITE DAMAGE TOLERANCE

Design values for residual strengths after BVID impact in tension and compression (as well as approximation for dent depth to impact energy).

Property	B-basis Value for all conditions
TAI ($\mu\epsilon$)	6900

Note: TAI estimated as $0.8 \cdot OHT$, and EKDF taken from PTS data.

$$E/t^2 = A \cdot (1 - e^{-B \cdot d_0/t})$$

Approximation model type: CDTM-T07

$$CAI = K_B \cdot EKDF \cdot CAI_{\infty-QI} \cdot (3 / K_{TOH})^C \cdot [1 + A / e^{B \cdot (E/t^2)}]$$

Property	Design Value
A (J/mm ²)	4.4
B	4.9

Property	Mean Value RT/Dry	B-basis K_B	E-KDF		
			70°C/W	90°C/W	120°C/W
$CAI_{\infty-QI}$ ($\mu\epsilon$)	2300	0.83	0.76	0.62	-
A	2.00				
B (mm ² /J)	0.5				
C	1.00				

...where:

- E is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- t the laminate thickness
- d_0 the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)

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- K_{TOH} is the Stress Concentration Factor for an infinite plate with a circular hole in by-pass load, calculated from laminate apparent moduli for X direction (*) as:

$$K_{TOH} = 1 + \{ 2 \cdot [(E_x/E_y)^{0.5} - \nu_{xy}] + E_x/G_{xy} \}^{0.5}$$

(*) CAI shall be calculated not only for the 0° direction (X parallel to material 0° reference), but also for the rest of ply orientations, i.e. the 45° and 90° directions, by rotating the X axis and re-evaluating previous expressions with the new values of E_x , E_y , G_{xy} and ν_{xy} . In a general case CAI strength shall be expressed as 3 scalar values with the allowable strains applicable to plies at 0° (CAI_0), at ±45° (CAI_{45}) and at 90° (CAI_{90}).

- Residual strength after transverse BVID near lightening holes

Property	B-basis value up to 70°C/W
CAI_{hole} (µε)	7600

...to be compared with the peak tangential strain calculated at the edge of a hole in a panel made of this material. Source: tests on impacted holed panels made of Hexcel HiTape/RTM6 UD-Tape.

- Residual strength after edge impact

No design data for CAI_E still available

Warning! Damage tolerance design values shall be considered preliminary and not used for structural justification assessments in certification/qualification processes (this material is still not qualified).